

DAY 1 / 30 · AZURE DATA ENGINEER SERIES

Azure Fundamentals

What Every Data Engineer Must Know — with real-time examples, tips & interview questions

✦ Follow for Day 2 → Azure Data Factory Deep Dive

1

What is Azure Data Factory (ADF)?

A cloud-based ETL and orchestration tool used to move and transform data between different sources.

◆ REAL-TIME EXAMPLE

Your company gets daily sales data from an on-prem SQL Server. ADF picks up that data every night, moves it to Azure Data Lake, transforms it (cleans/filters), and loads it into Azure Synapse for reporting — all on a schedule, without manual work.

💡 TIP

Always relate ADF to "automation of data movement."
Interviewers like when you connect it to a business use case, not just a definition.

2 ADLS Gen1 vs Gen2

Gen2 is Gen1 + Blob Storage capabilities — it supports a hierarchical namespace (folders, like a file system) and is cheaper, faster, and more secure.

◆ REAL-TIME EXAMPLE

If you're storing millions of log files from an e-commerce site, Gen2 lets you organize them into folders like /2026/July/13/ instead of one flat container — making retrieval much faster.

💡 TIP

Mention "hierarchical namespace" — it's the #1 keyword interviewers listen for.

3 What is a Resource Group?

A logical container that holds related Azure resources (storage, databases, pipelines) for a project.

◆ REAL-TIME EXAMPLE

For a "Customer Analytics" project, you'd group ADF, ADLS, and Synapse together in one Resource Group — so you can manage, monitor, and even delete them all together instead of hunting resources individually.

💡 TIP

Say "it helps with cost tracking and access management" — this shows you understand real-world usage, not just theory.

4

Structured vs Semi-structured vs Unstructured

Structured = tables (SQL databases) · Semi-structured = JSON/XML · Unstructured = images, videos, PDFs.

◆ REAL-TIME EXAMPLE

A retail company has structured sales data in SQL, semi-structured clickstream logs in JSON from their website, and unstructured product images stored in Blob Storage — Azure handles all three differently.

💡 TIP

Give one-line examples of each type — interviewers love quick, concrete answers over long explanations.

5 PaaS vs IaaS vs SaaS

IaaS = you manage the OS/infra (VMs) · PaaS = Azure manages infra, you manage app/data (like ADF, Azure SQL DB) · SaaS = fully managed app (like Office 365).

◆ REAL-TIME EXAMPLE

Using an Azure VM to host your own database = IaaS. Using Azure SQL Database (no server management needed) = PaaS. Using Outlook online = SaaS.

💡 TIP

Use the "who manages what" mental checklist — infra, runtime, app, data — it helps you answer under pressure.

Bonus Interview Questions

- ? Difference between Azure Blob Storage and ADLS?
- ? How does ADF differ from SSIS?
- ? What are Linked Services in ADF?
- ? What are Integration Runtimes, and how many types exist?
- ? What is the difference between a Data Lake and a Data Warehouse?

Communication Tips — Practice Daily

Technical knowledge gets you the interview. Communication gets you the offer.

- 1 **Structure your answers:** Definition → Real-world example → Why it matters. Never just recite theory.
- 2 **Use the STAR method** (Situation, Task, Action, Result) for scenario-based questions.
- 3 **Think out loud** if unsure — interviewers value reasoning over silence.
- 4 **Keep answers under 2 minutes.** Be crisp, then ask "Would you like me to go deeper?"
- 5 **Simplify for a non-technical listener.** If you can explain ADF to your mom, you truly understand it.
- 6 **End with confidence.** Avoid trailing off — close strong: "That's how I'd approach it."
- 7 **Do mock interviews out loud** — speaking clarity matters as much as technical knowledge.

Let's Grow Together

★ **Follow** me for Day 2 tomorrow — Azure Data Factory in depth (pipelines, activities, triggers).

🔄 **Repost/share** if you know someone preparing for Azure Data Engineer interviews.

💬 **Comment "Azure"** below if you'd like to be part of this journey.